CYBERSECURITY FUNDAMENTALS

Complete 50-Page Course Textbook

Beginner → Intermediate

Theory • Defensive Practices • Case Studies • Exercises • Assessments

A structured learning resource for understanding, preventing, detecting, and responding to cybersecurity risks.

Course Overview

Purpose: This course introduces the principles and defensive practices used to protect computers, networks, applications, accounts, and data. It explains common threats, security controls, risk management, incident response, and safe security operations.

Learning outcomes: Learners should be able to explain core cybersecurity concepts, identify common threats, assess basic risks, apply account and network security practices, understand encryption and access control, recognize social engineering, plan basic incident response, and describe defensive security roles.

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Lesson 1: Introduction to Cybersecurity

Definition

Cybersecurity is the practice of protecting information systems, networks, devices, applications, and data from unauthorized access, misuse, disruption, alteration, or destruction.

Why it matters

Modern education, banking, healthcare, government, communication, and business depend on digital systems. A security failure can affect confidentiality, finances, operations, safety, and trust.

Security objectives

Security programs generally seek to protect confidentiality, integrity, and availability while also supporting authenticity, accountability, privacy, and resilience.

Defensive mindset

Cybersecurity is not only about tools. People, processes, technology, governance, training, monitoring, and recovery all contribute to security.

Exercise

List five digital assets you use in daily life and one security concern for each.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 2: Cybersecurity Terminology

Asset

An asset is something valuable that needs protection, such as data, devices, accounts, applications, services, or reputation.

Threat

A threat is a potential cause of harm. Threats can be accidental, environmental, technical, or malicious.

Vulnerability

A vulnerability is a weakness that could be exploited or otherwise lead to harm.

Risk

Risk describes the possibility and impact of an undesirable event. Risk management helps organizations prioritize limited resources.

Control

A security control is a safeguard used to prevent, detect, deter, or respond to threats.

Exercise

For a school database, identify three assets, three threats, three vulnerabilities, and three controls.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 3: CIA Triad

Confidentiality

Confidentiality means information is accessible only to authorized people or systems. Access controls, encryption, and appropriate data handling support confidentiality.

Integrity

Integrity means information remains accurate, complete, and protected against unauthorized alteration.

Availability

Availability means authorized users can access systems and information when needed.

Balancing objectives

Controls can affect more than one objective. For example, strong authentication can protect confidentiality, while backups primarily support availability and recovery.

Exercise

Give two examples of a confidentiality incident, two integrity incidents, and two availability incidents.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 4: Assets, Threats, Vulnerabilities and Risk

Risk relationship

A simplified model is that risk is influenced by the likelihood of an event and its potential impact. More detailed assessments consider assets, threat actors, vulnerabilities, existing controls, and uncertainty.

Risk assessment

Identify assets, threats, vulnerabilities, consequences, likelihood, and current controls. Then prioritize treatment.

Risk treatment

Common approaches include reducing risk with controls, avoiding certain activities, transferring some financial impact, or accepting risk when justified.

Defense in depth

Use multiple layers of controls so failure of one control does not automatically result in compromise.

Exercise

Create a small risk register for a student information system with five risks and suggested controls.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 5: Malware

Definition

Malware is malicious software designed to perform unauthorized or harmful actions.

Major categories

Common categories include viruses, worms, trojans, ransomware, spyware, and botnet-related malware. Categories can overlap in real incidents.

Ransomware

Ransomware can prevent access to data or systems and may involve data theft. Defenses include backups, patching, least privilege, endpoint protection, segmentation, and user awareness.

Defense

Use reputable security software, keep systems updated, restrict privileges, maintain tested backups, and avoid untrusted files and links.

Exercise

Create a defensive checklist for reducing malware risk on a school computer lab.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 6: Phishing and Social Engineering

Social engineering

Social engineering manipulates people into revealing information, performing actions, or bypassing normal security procedures.

Phishing

Phishing commonly uses deceptive messages or websites to trick recipients. Variants may target specific individuals or organizations.

Warning signs

Look for unexpected requests, urgency, unusual payment instructions, suspicious links, mismatched sender information, unexpected attachments, and requests for sensitive information.

Safe response

Verify important requests through an independent trusted channel. Do not disclose credentials or sensitive information simply because a message appears urgent.

Exercise

Write a checklist students can use to evaluate suspicious email or messaging requests.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 7: Passwords and Authentication

Authentication

Authentication verifies that a person or system is who or what it claims to be.

Password principles

Use long, unique passwords or passphrases for important accounts. Avoid predictable personal information and password reuse.

Password managers

A reputable password manager can generate and store unique credentials, reducing the need to remember many passwords.

Credential protection

Never share passwords unnecessarily. Be cautious of requests for passwords or authentication codes, even when the request appears to come from an authority.

Exercise

Design a school password policy covering uniqueness, password storage, account recovery, and handling of compromised credentials.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 8: Multi-Factor Authentication

MFA

Multi-factor authentication requires two or more different categories of authentication evidence, such as something you know, have, or are.

Why MFA helps

A stolen password alone is less useful to an attacker when an additional independent factor is required.

Implementation

Organizations should provide secure enrollment, recovery procedures, device management, and user education.

Recovery

Recovery methods are part of the security design. Weak recovery can undermine otherwise strong authentication.

Exercise

Explain why MFA is valuable for student, teacher, and administrator accounts.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 9: Access Control

Least privilege

Give users and systems only the permissions necessary to perform their tasks.

Role-based access

Role-based access control assigns permissions according to job or responsibility rather than individually granting every permission.

Separation of duties

Important actions can be divided among roles so that one person or account does not have excessive control.

Account lifecycle

Create, modify, review, disable, and remove accounts according to defined procedures.

Exercise

Define appropriate permissions for student, teacher, administrator, and support roles in an e-learning system.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 10: Network Security

Network security

Network security protects communication infrastructure, devices, services, and traffic from unauthorized access and disruption.

Segmentation

Segmentation separates systems or users into logical zones so that compromise in one area has less opportunity to spread.

Secure protocols

Use appropriately secured protocols and avoid transmitting sensitive information through insecure channels.

Monitoring

Network monitoring can help identify unusual traffic, failures, and potential security incidents.

Exercise

Design a basic secure network layout for a school containing students, teachers, administrators, servers, and guest devices.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 11: Firewalls

Definition

A firewall controls network traffic according to defined rules. It can operate at network boundaries or on individual endpoints.

Default-deny concept

Where appropriate, a restrictive policy can block traffic unless it is explicitly allowed. Rules must be carefully designed to avoid breaking legitimate services.

Rule management

Document rules, review them periodically, remove obsolete access, and restrict administrative access.

Limitations

A firewall is not a complete security solution. It does not replace patching, authentication, endpoint protection, backups, monitoring, or user awareness.

Exercise

Explain three legitimate uses of a firewall and three security tasks a firewall cannot solve by itself.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 12: Secure Wi-Fi

Wireless risks

Poorly configured wireless networks can expose users to unauthorized access, weak authentication, or insecure communication.

Defensive practices

Use modern security settings supported by the equipment, strong administrator credentials, updated firmware, separate guest access, and appropriate network segmentation.

Guest networks

Guest devices should generally be separated from sensitive internal systems when feasible.

Administration

Change default administrative credentials and restrict management access.

Exercise

Create a secure Wi-Fi checklist for a small school or office.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 13: Encryption and Cryptography

Encryption

Encryption transforms readable information into a protected form so unauthorized parties cannot easily understand it without the appropriate key.

Data in transit and at rest

Data may need protection while moving across networks and while stored on devices or servers.

Symmetric encryption

Symmetric cryptography uses the same secret key for encryption and decryption.

Asymmetric cryptography

Asymmetric cryptography uses a related public/private key pair and supports applications such as secure key exchange and digital signatures.

Hashing

Hash functions produce fixed-length representations of data and are commonly used for integrity checks and secure password-storage schemes when implemented with appropriate password-hashing algorithms.

Exercise

Explain the difference between encryption and hashing and give one appropriate use for each.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 14: Web Security

Web threats

Web applications can face risks such as injection, broken access control, insecure authentication, cross-site scripting, insecure configuration, and data exposure.

Secure sessions

Use secure session management, appropriate cookie protections, timeouts, and protection of authentication state.

Input handling

Treat user input as untrusted. Validate it appropriately and use safe APIs or parameterized mechanisms rather than constructing dangerous queries or commands.

Transport security

Use properly configured encrypted connections for sensitive web traffic.

Exercise

Create a defensive checklist for securing an online course website.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 15: Application Security

Secure development

Application security should be considered throughout requirements, design, coding, testing, deployment, and maintenance.

Threat modeling

Identify valuable assets, trust boundaries, likely threats, misuse cases, and security controls before implementation.

Dependency security

Keep libraries and frameworks maintained, inventory dependencies, and monitor for relevant security advisories.

Secrets

Do not hard-code passwords, API keys, or other sensitive credentials into source code or public repositories.

Exercise

Create five application-security requirements for a Django-based e-learning platform.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 16: Operating System Security

Patching

Security updates address known weaknesses. Establish a process for testing and deploying updates according to risk and operational requirements.

Least privilege

Users should not routinely operate with administrative rights unless required.

Host firewall

Endpoint firewalls can restrict unnecessary inbound and outbound communication.

Endpoint protection

Security software, logging, configuration management, and device control can improve protection.

Exercise

Create a 10-item workstation hardening checklist for students and staff.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 17: Mobile Device Security

Mobile risks

Mobile devices contain accounts, communications, files, and applications and may be lost or stolen.

Defensive controls

Use device locks, updates, reputable applications, encryption where available, and account protection.

Application permissions

Review permissions and avoid granting unnecessary access to sensitive resources.

Lost devices

Use appropriate device-finding, remote-management, or remote-wipe capabilities when available and configured beforehand.

Exercise

Write a mobile security checklist for students using phones for online learning.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 18: Cloud Security

Shared responsibility

Cloud security responsibilities are divided between the provider and customer according to the service model. Customers remain responsible for correct configuration of their own resources and data.

Identity

Strong authentication and least privilege are critical because cloud resources are often managed through centralized identities.

Configuration

Misconfigured storage, permissions, networks, and logging can expose resources.

Monitoring

Use available audit logs and security monitoring to detect unusual activity.

Exercise

List five security checks for a cloud-hosted e-learning application.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 19: Database Security

Database assets

Databases may contain identities, grades, course content, payment-related information, or other sensitive records.

Access control

Applications should access databases using accounts with only the permissions required.

Secure queries

Use safe database interfaces and parameterized queries to reduce injection risks.

Backups

Maintain protected, tested backups and control who can access them.

Exercise

Create a database-security checklist for student records.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 20: Data Protection and Privacy

Data classification

Organizations can classify information according to sensitivity and business or legal requirements.

Data minimization

Collect and retain only information needed for legitimate purposes.

Protection

Use appropriate access control, encryption, retention, secure disposal, and monitoring.

Privacy by design

Consider privacy during system design rather than adding privacy controls only after deployment.

Exercise

Classify ten example data items in an e-learning system as public, internal, confidential, or highly sensitive and explain your reasoning.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 21: Security Policies

Purpose

Security policies establish expected behavior, responsibilities, and minimum security requirements.

Examples

Acceptable-use, password, access-control, incident-response, backup, device, remote-work, and data-handling policies are common.

Policy lifecycle

Policies should be approved, communicated, implemented, reviewed, and updated.

Enforcement

A policy is useful only when supported by procedures, training, technical controls, and accountability.

Exercise

Draft the headings for a cybersecurity policy for a small educational institution.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 22: Security Awareness

Human factor

People are part of the security system. Awareness reduces preventable errors but should not be used to blame victims of sophisticated attacks.

Training topics

Teach phishing recognition, password safety, MFA, device security, reporting procedures, data handling, and safe use of external services.

Reporting culture

Users should know how to report suspicious messages or incidents quickly and without fear of inappropriate punishment.

Continuous learning

Threats and technologies change, so awareness should be refreshed periodically.

Exercise

Create a one-page security awareness campaign for students.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 23: Vulnerability Management

Vulnerability

A vulnerability is a weakness that could contribute to a security incident.

Inventory

You cannot reliably protect unknown assets. Maintain an inventory of devices, applications, services, and important dependencies.

Scanning

Defensive vulnerability scanning can identify known weaknesses for remediation. Results require validation because automated tools can produce false positives or miss context.

Remediation

Prioritize based on severity, exposure, asset importance, exploitability, and business impact.

Exercise

Create a monthly vulnerability-management workflow for a school network.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 24: Security Monitoring

Monitoring

Security monitoring collects and analyzes relevant events to identify failures, suspicious activity, and possible incidents.

Logs

Useful logs may come from authentication systems, endpoints, servers, applications, network devices, and cloud services.

Baseline

Understanding normal behavior helps teams recognize anomalies.

Alert quality

Too many low-value alerts can overwhelm analysts. Prioritize actionable signals and investigate important deviations.

Exercise

List ten useful security events to monitor on an e-learning platform.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 25: Incident Response

Incident

A security incident is an event that threatens or violates security policies or controls and requires investigation or response.

Preparation

Maintain contacts, procedures, logging, backups, tools, responsibilities, and communication plans before incidents occur.

Detection and analysis

Identify what happened, what systems are affected, and the likely scope. Preserve relevant evidence.

Containment and recovery

Limit further harm, remove the cause where possible, restore services safely, and monitor for recurrence.

Lessons learned

After recovery, document causes, impact, response performance, and improvements.

Exercise

Create a defensive incident-response plan for a suspected compromised student account.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 26: Digital Forensics Fundamentals

Forensics

Digital forensics is the disciplined examination of digital evidence to understand events while preserving evidence integrity.

Evidence handling

Document collection, access, storage, and analysis. Preserve originals and use appropriate procedures.

Artifacts

Potential evidence may include logs, timestamps, system records, application data, and network records.

Chain of custody

A chain of custody records who handled evidence, when, why, and how it was transferred.

Exercise

Explain why evidence preservation and documentation matter during a cybersecurity investigation.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 27: Backup and Disaster Recovery

Backups

Backups provide copies of data that can be restored after deletion, corruption, hardware failure, or security incidents.

Backup strategy

Consider frequency, retention, storage location, access control, encryption, and restoration testing.

Recovery objectives

Recovery Time Objective (RTO) describes a target for restoring service. Recovery Point Objective (RPO) describes how much recent data loss may be acceptable.

Testing

A backup that has never been restored should not be assumed to work.

Exercise

Design a backup plan for an e-learning database and course-content repository.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 28: Secure Software Development

Security throughout SDLC

Security should appear in requirements, architecture, implementation, testing, deployment, and maintenance.

Code review

Peer review can identify logic errors, insecure assumptions, exposed secrets, and poor handling of sensitive data.

Testing

Security testing may include dependency checks, static analysis, dynamic testing, configuration review, and appropriate authorized penetration testing.

Secure defaults

Applications should ship with safe configurations and avoid unnecessary exposure.

Exercise

Create a secure-development checklist for a small web application.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 29: Cybersecurity Risk Management

Risk program

Risk management connects technical security decisions to organizational objectives.

Risk register

A risk register records risks, causes, affected assets, likelihood, impact, existing controls, owners, and treatment decisions.

Prioritization

High-impact risks affecting critical assets usually deserve greater attention, especially when realistic attack paths exist.

Continuous review

Risk changes as systems, threats, business processes, and controls change.

Exercise

Create five entries for a cybersecurity risk register for a college.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 30: Security Architecture

Architecture

Security architecture organizes technical and procedural controls across identities, networks, applications, endpoints, data, and monitoring.

Zero Trust concept

Zero Trust emphasizes continuous verification and least-privilege access rather than assuming that network location alone makes a user or device trustworthy.

Defense in depth

Use multiple independent or complementary controls across layers.

Segmentation

Separate sensitive systems and limit unnecessary communication paths.

Exercise

Draw a high-level security architecture for an e-learning system and label identity, application, database, network, logging, and backup controls.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 31: Email and Messaging Security

Threats

Email and messaging can carry phishing, malicious attachments, impersonation, data leakage, and fraudulent requests.

Verification

For sensitive requests, verify through an independent channel rather than relying on the same message.

Attachments and links

Treat unexpected files and links cautiously. Use security controls and scanning where available.

Account protection

MFA, strong authentication, monitoring, and recovery procedures reduce account takeover risk.

Exercise

Create a checklist for deciding whether an unexpected school email should be trusted, reported, or ignored.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 32: Endpoint Security

Endpoints

Endpoints include laptops, desktops, phones, tablets, and other user-accessible devices.

Endpoint controls

Useful controls include patching, malware protection, encryption, device management, firewall configuration, application control, and logging.

Configuration

Standard secure configurations reduce unnecessary exposure and make monitoring easier.

Remote work

Remote devices need the same basic security principles even when they are outside the organization's physical network.

Exercise

Build a 12-point endpoint-security checklist.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 33: Identity and Access Management

IAM

Identity and Access Management controls who can access resources and under what conditions.

Identity lifecycle

Provision identities, assign appropriate roles, review access, and remove access promptly when it is no longer needed.

Privileged access

Administrative accounts require stronger controls because compromise can have broad consequences.

Access reviews

Periodic reviews help detect excessive or outdated permissions.

Exercise

Design an access lifecycle for student and teacher accounts from registration through account closure.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 34: Security Operations

Security operations

Security operations coordinates monitoring, investigation, incident response, vulnerability management, and continuous improvement.

SOC concept

A Security Operations Center (SOC) may centralize alert monitoring and analysis. Smaller organizations may distribute these functions across IT and security staff.

Playbooks

Playbooks provide structured response steps for common events such as suspicious login, malware alert, or exposed credential.

Metrics

Useful metrics include alert volume, response time, remediation time, patch coverage, MFA coverage, and recurring incident types.

Exercise

Create three defensive playbooks: suspicious login, phishing report, and malware alert.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 35: Cybersecurity Careers

Security roles

Career paths include security analyst, security engineer, incident responder, GRC specialist, security administrator, cloud security specialist, application security specialist, digital forensics analyst, and security architect.

Core skills

Communication, networking, operating systems, identity, risk analysis, documentation, scripting, troubleshooting, and continuous learning are valuable.

Ethics

Security professionals must respect authorization, privacy, laws, policies, and scope. Testing systems without permission can cause harm and legal problems.

Learning path

Start with fundamentals, build safe lab projects, document defensive work, learn networking and operating systems, and develop specialization gradually.

Exercise

Choose three cybersecurity roles and list the skills each requires.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 36: Defensive Capstone Project

Project brief

Design a cybersecurity improvement plan for a fictional educational institution with an e-learning platform, staff accounts, student accounts, Wi-Fi, web servers, and a database.

Required deliverables

Asset inventory, threat model, risk register, security architecture, account policy, network-security plan, backup plan, incident-response plan, awareness plan, and monitoring checklist.

Scenario

The institution has experienced repeated phishing reports, outdated devices, shared passwords, weak backup procedures, and unclear incident-reporting responsibilities.

Expected outcome

Propose a prioritized defensive roadmap. Explain which risks are most important, which controls address them, who owns each action, and how success will be measured.

Exercise

Prepare a five-page capstone report and a short presentation explaining the plan.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Lesson 37: Revision Guide

Review

Revise CIA triad, threat/vulnerability/risk, malware, phishing, authentication, access control, network security, encryption, application security, incident response, backups, and security governance.

Defense mindset

Ask four questions: What are we protecting? What could go wrong? Which controls reduce the risk? How will we know the controls work?

Checklist

Review accounts, permissions, updates, backups, logging, network segmentation, endpoint controls, security awareness, and incident procedures.

Professional practice

Document decisions, stay within authorization, protect sensitive information, and continually improve.

Exercise

Explain a complete defensive security program for a small e-learning organization in one page.

Lesson checkpoint: Write three key ideas from this lesson, one question you still have, and one defensive action you could apply.

Defensive Case Study: Securing an E-Learning Platform

Scenario: A fictional school operates an online learning platform with student and teacher accounts, a web application, a database, Wi-Fi, cloud backups, and administrative systems. The school has experienced phishing attempts and several outdated devices.

Step 1 — Identify assets

List accounts, student records, course content, application servers, databases, network equipment, backup repositories, and administrative systems. Classify them according to importance and sensitivity.

Step 2 — Identify risks

Consider account compromise, malware, data exposure, service disruption, unauthorized changes, weak backups, and excessive privileges.

Step 3 — Select controls

Prioritize MFA, least privilege, patching, endpoint protection, secure configuration, network segmentation, protected backups, logging, awareness training, and incident-response procedures.

Step 4 — Monitor and improve

Define useful security events and metrics. Review access, vulnerabilities, backups, and incidents regularly. Update controls as risks change.

Final Assessment: 60 Questions and Answers

Attempt each question before reading the answer.

1. What is cybersecurity?

Answer: The practice of protecting systems, networks, devices, applications, and data from unauthorized or harmful activity.

2. What is an asset?

Answer: Something valuable that requires protection.

3. What is a threat?

Answer: A potential cause of harm.

4. What is a vulnerability?

Answer: A weakness that could contribute to a security incident.

5. What is risk?

Answer: The possibility and impact of an undesirable event.

6. What is a security control?

Answer: A safeguard used to prevent, detect, deter, or respond to security risks.

7. What does CIA stand for?

Answer: Confidentiality, Integrity, and Availability.

8. What is confidentiality?

Answer: Preventing unauthorized access to information.

9. What is integrity?

Answer: Protecting information from unauthorized or improper alteration.

10. What is availability?

Answer: Ensuring authorized access when needed.

11. What is malware?

Answer: Malicious software intended to perform unauthorized or harmful actions.

12. Name three malware categories.

Answer: Ransomware, spyware, and trojans.

13. What is phishing?

Answer: A deceptive attempt to trick people into revealing information or performing unsafe actions.

14. What is social engineering?

Answer: Manipulating people to bypass normal security behavior or reveal information.

15. Why use unique passwords?

Answer: To reduce the impact of one compromised password on other accounts.

16. What is MFA?

Answer: Authentication using two or more different factors.

17. What is least privilege?

Answer: Giving only the permissions necessary for a task.

18. What is role-based access control?

Answer: Assigning permissions according to defined roles.

19. What is network segmentation?

Answer: Separating networks or systems to limit unnecessary access and potential spread.

20. What is a firewall?

Answer: A control that filters network traffic according to rules.

21. Why secure Wi-Fi?

Answer: To prevent unauthorized access and protect network communications.

22. What is encryption?

Answer: Transforming data into protected form using a cryptographic method and key.

23. What is symmetric encryption?

Answer: Encryption using the same secret key for encryption and decryption.

24. What is asymmetric cryptography?

Answer: Cryptography using related public and private keys.

25. What is hashing?

Answer: A one-way transformation that produces a fixed-length digest of data.

26. Why patch systems?

Answer: To address known vulnerabilities and improve security.

27. What is web security?

Answer: Protecting web applications, services, sessions, data, and users from relevant threats.

28. Why is input validation important?

Answer: It helps ensure application input is handled safely and as expected.

29. What is a security policy?

Answer: A formal statement of security requirements and expected behavior.

30. What is security awareness?

Answer: Education that helps people recognize and respond appropriately to security risks.

31. What is vulnerability management?

Answer: Identifying, prioritizing, remediating, and tracking security weaknesses.

32. What is security monitoring?

Answer: Collecting and analyzing security-relevant events to identify problems or suspicious activity.

33. What is incident response?

Answer: A structured process for detecting, containing, investigating, recovering from, and learning from security incidents.

34. Why preserve evidence?

Answer: To maintain its reliability and support accurate investigation.

35. What is chain of custody?

Answer: Documentation of who handled evidence, when, why, and how.

36. Why are backups important?

Answer: They allow restoration after loss, corruption, hardware failure, or certain security incidents.

37. What is RTO?

Answer: Recovery Time Objective: a target for restoring a service.

38. What is RPO?

Answer: Recovery Point Objective: the amount of recent data loss that may be acceptable.

39. What is secure software development?

Answer: Integrating security into requirements, design, coding, testing, deployment, and maintenance.

40. What is threat modeling?

Answer: Systematically identifying assets, threats, trust boundaries, and controls during design.

41. Why protect secrets?

Answer: Exposed credentials can provide unauthorized access to systems and data.

42. What is cloud shared responsibility?

Answer: Security responsibilities are divided between the provider and customer according to the service and configuration.

43. Why protect databases?

Answer: They often contain valuable or sensitive records.

44. What is data minimization?

Answer: Collecting and retaining only information necessary for legitimate purposes.

45. What is defense in depth?

Answer: Using multiple layers of complementary security controls.

46. What is Zero Trust?

Answer: A model emphasizing verification and least privilege rather than implicit trust.

47. What is endpoint security?

Answer: Protecting user devices such as computers and phones.

48. What is IAM?

Answer: Identity and Access Management.

49. Why review permissions?

Answer: To detect excessive, outdated, or inappropriate access.

50. What is a SOC?

Answer: A Security Operations Center that may coordinate security monitoring and response.

51. What is a playbook?

Answer: A documented set of response steps for a recurring security event.

52. Why is security training important?

Answer: It helps users recognize risks and follow safer procedures.

53. What is a dark pattern?

Answer: A deceptive interface that manipulates users into unintended actions.

54. Why is privacy part of security?

Answer: Security controls help protect personal information from unauthorized exposure or misuse.

55. What is a risk register?

Answer: A structured record of identified risks, impacts, controls, owners, and treatment decisions.

56. What is an asset inventory?

Answer: A record of systems, devices, applications, and other assets that need protection.

57. Why is logging important?

Answer: Logs provide information for monitoring, investigation, troubleshooting, and accountability.

58. Why test backups?

Answer: To verify that data can actually be restored when needed.

59. What is the most important rule for security testing?

Answer: Only test systems when you have explicit authorization and remain within the defined scope.

Capstone Project Rubric

Category	Excellent	Good	Needs Improvement
Asset/Risk Analysis	Comprehensive and prioritized	Mostly complete	Major assets or risks missing
Controls	Layered, practical, justified	Useful controls	Controls poorly matched to risks
Identity	Strong MFA, least privilege, lifecycle	Basic access controls	Excessive or unclear access
Network	Segmentation and secure configuration	Basic network controls	Major exposure remains
Data	Classification, protection, backups	Basic protection	Weak protection or recovery
Monitoring	Clear events, alerts, response	Some monitoring	Little visibility
Incident Response	Complete preparation-to-recovery plan	Major stages covered	Important stages missing
Awareness	Relevant training and reporting	Basic training	No clear awareness plan
Documentation	Clear, organized, actionable	Mostly clear	Difficult to use

Cybersecurity Glossary

Asset: A valuable resource that needs protection.

Authentication: Verifying an identity.

Authorization: Determining what an authenticated identity is allowed to do.

Availability: Authorized access to systems and information when needed.

Backup: A copy of data maintained for restoration.

CIA Triad: Confidentiality, Integrity, and Availability.

Control: A safeguard that reduces or manages risk.

Encryption: Cryptographic protection that makes data unreadable without appropriate keys.

Firewall: A control that filters network traffic according to rules.

Hash: A fixed-length digest produced by a hash function.

Incident: An event that threatens or violates security.

Integrity: Protection against unauthorized or improper alteration.

Least privilege: Minimum necessary access.

Malware: Malicious software.

MFA: Multi-factor authentication.

Phishing: Deceptive communication intended to manipulate a recipient.

Risk: Potential for harm considering likelihood and impact.

Threat: Potential cause of harm.

Vulnerability: A weakness that may contribute to harm.

Zero Trust: A model emphasizing verification and least privilege rather than implicit trust.

End of Cybersecurity Fundamentals

Cybersecurity is a continuous defensive discipline. Strong security combines people, processes, technology, governance, monitoring, and recovery. Continue learning through authorized labs, documented projects, and responsible security practice.